

Name: Saiprasad Ashokrao Kawdikar

PRN No: 202501040030

4.2.5. Titanic Dataset Analysis and Data Cleaning

01:36

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

Note: Refer to the visible test case for better reference.

```
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# 1. Display the first 5 rows of the dataset
print(data.head())

# 2. Display the last 5 rows of the dataset
print(data.tail())

# 3. Get the shape of the dataset
print(data.shape)

# 4. Get a summary of the dataset (info)
print(data.info())

# 5. Get basic statistics of the dataset
print(data.describe())

# 6. Check for missing values
print(data.isnull().sum())

# 7. Fill missing values in the 'Age' column with the median age
data['Age'] = data['Age'].fillna(data['Age'].median())
```

8. Fill missing values in the 'Embarked' column with the mode

```
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

9. Drop the 'Cabin' column due to many missing values

```
data = data.drop(columns=['Cabin'])
```

10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'

```
data['FamilySize'] = data['SibSp'] + data['Parch']
```

Explorer

Average time **0.166 s** 166.00 ms Maximum time **0.166 s** 166.00 ms ✓ 1 out of 1 shown test case(s) passed

✓ Test case 1 166 ms Debug

Expected output	Actual output
PassengerId Survived Pclass Fare Cabin Embarked	PassengerId Survived Pclass Fare Cabin Embarked
0 1 0 3 7.2500 NaN S	0 1 0 3 7.2500 NaN S
1 2 1 1 71.2833 C85 C	1 2 1 1 71.2833 C85 C
2 3 1 3 7.9250 NaN S	2 3 1 3 7.9250 NaN S
3 4 1 1 53.1000 C123 S	3 4 1 1 53.1000 C123 S
4 5 0 3 8.0500 NaN S	4 5 0 3 8.0500 NaN S
[5 rows x 12 columns]	[5 rows x 12 columns]
PassengerId Survived Pclass Fare Cabin Embarked	PassengerId Survived Pclass Fare Cabin Embarked
886 887 0 2 13.00 NaN S	886 887 0 2 13.00 NaN S
887 888 1 1 30.00 B42 S	887 888 1 1 30.00 B42 S
888 889 0 3 23.45 NaN S	888 889 0 3 23.45 NaN S
889 890 1 1 30.00 C148 C	889 890 1 1 30.00 C148 C
890 891 0 3 7.75 NaN Q	890 891 0 3 7.75 NaN Q

Average time

0.166 s

166.00 ms



Maximum time

0.166 s

166.00 ms

**1 out of 1 shown test case(s) passed**

```
[5 rows x 12 columns]
(891, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  --
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass           891 non-null    int64
3   Name             891 non-null    object
4   Sex              891 non-null    object
5   Age              714 non-null    float64
6   SibSp            891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket           891 non-null    object
9   Fare             891 non-null    float64
10  Cabin            204 non-null    object
11  Embarked         889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB
None
.....PassengerId.....Survived.....Pclass.....SibSp.....Parch.....Fare
count.....891.000000.....891.000000.....891.000000.....891.000000.....891.000000.....891.000000
mean.....446.000000.....0.383838.....2.308642.....0.523008.....0.381594.....32.204208
std.....257.353842.....0.486592.....0.836071.....1.102743.....0.806057.....49.693429

[5 rows x 12 columns]
(891, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  --
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass           891 non-null    int64
3   Name             891 non-null    object
4   Sex              891 non-null    object
5   Age              714 non-null    float64
6   SibSp            891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket           891 non-null    object
9   Fare             891 non-null    float64
10  Cabin            204 non-null    object
11  Embarked         889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB
None
.....PassengerId.....Survived.....Pclass.....SibSp.....Parch.....Fare
count.....891.000000.....891.000000.....891.000000.....891.000000.....891.000000.....891.000000
mean.....446.000000.....0.383838.....2.308642.....0.523008.....0.381594.....32.204208
std.....257.353842.....0.486592.....0.836071.....1.102743.....0.806057.....49.693429
```

Terminal

Test cases

